

ECON 3350/7350 Applied Econometrics for Macroeconomics and Finance

Lecture 9: VAR Models (1)

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VAR Models

Recommended readings

Author	Title	Chapter	Call No
Enders	Applied Econometric Time Series, 4e	5	HB139 .E55 2015
Brooks	Introductory Econometrics for Finance, 3e	7	HG173 .B76 2014
Verbeek	A Guide to Modern Econometrics	9.1, 9.4	HB139 .V465 2012

Multivariate Time Series

Multiple equations models involve several variables as well as several equations.

Multiple equations models imply there are more than one **endogenous variable**.

Single equation models involve only one endogenous variable: the **dependent** variable.

Multivariate Time Series

Endogenous Variables: variables which are explained by other variables in the system. Another name for endogenous variable is **dependent** variable.

Exogenous Variables: variables which are thought to be exogenous to the system, either by theoretical assumption or by model definition. Other names for exogenous variables include **predictor** variables and **independent** variables.

Multiple Equation Models (MEM) have as many equations as endogenous variables.

Classification of MEMs

MEMs can be broadly classified in two main categories:

Simultaneous Equations Models (SEM): two or more endogenous variables are jointly determined and there is feedback (i.e., information flow in both directions) between them.

- Examples: price and the quantity demanded in a given time period are negatively related and there is feedback; Structural VAR Models (SVAR).

Non-simultaneous Equations Models: “reduced form” models derived from SEMs.

- Examples: Seemingly Unrelated Regressions Models (SUR), Vector Autoregressive Models (VAR), Vector ARMA Models (VARMA), Vector Error Correction Models (VEC), VAR with exogenous variables (VARX).

Vector Autoregressive Models (VARs)

VARs were popularized in economics by Chris Sims in 1980. These reduced form models are a generalization of the univariate AR models.

VAR models are characterized by treating all variables as endogenous in the system.

VARs are special cases of VARMA, which are generalizations of the univariate ARMA models.

VARs are extremely easy to work with—many extensions created in the past 30+ years:

- VECM: An extension of the single equation ECM form of an ARDL model to the multivariate case.
- VARX: One or more exogenous variables are included in the model.
- Also, time-varying parameter VARs, regime-switching VARs, factor-augmented VARs, among many others.

Structural VAR (SVAR)

Recall the ARDL(1, 1) model:

$$y_t = b_{10} - b_{12}z_t + \gamma_{11}y_{t-1} + \gamma_{12}z_{t-1} + \varepsilon_{yt}.$$

Suppose we can also formulate a similar model with z_t as the dependent variable:

$$z_t = b_{20} - b_{21}y_t + \gamma_{21}y_{t-1} + \gamma_{22}z_{t-1} + \varepsilon_{zt}.$$

Assume that ε_{yt} and ε_{zt} are **uncorrelated** white noise processes, i.e.

$$E \left[\begin{pmatrix} \varepsilon_{yt} \\ \varepsilon_{zt} \end{pmatrix} \right] = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \text{Var} \left[\begin{pmatrix} \varepsilon_{yt} \\ \varepsilon_{zt} \end{pmatrix} \right] = \begin{pmatrix} \sigma_y^2 & 0 \\ 0 & \sigma_z^2 \end{pmatrix}.$$

SVAR

Write the system in matrix form:

$$\underbrace{\begin{pmatrix} 1 & b_{12} \\ b_{21} & 1 \end{pmatrix}}_{\mathbf{B}} \underbrace{\begin{pmatrix} y_t \\ z_t \end{pmatrix}}_{\mathbf{x}_t} = \underbrace{\begin{pmatrix} b_{10} \\ b_{20} \end{pmatrix}}_{\gamma_0} + \underbrace{\begin{pmatrix} \gamma_{11} & \gamma_{12} \\ \gamma_{21} & \gamma_{22} \end{pmatrix}}_{\mathbf{\Gamma}_1} \underbrace{\begin{pmatrix} y_{t-1} \\ z_{t-1} \end{pmatrix}}_{\mathbf{x}_{t-1}} + \underbrace{\begin{pmatrix} \varepsilon_{yt} \\ \varepsilon_{zt} \end{pmatrix}}_{\boldsymbol{\varepsilon}_t}.$$

This is referred to as the **structural** VAR.

Now, let

$$\mathbf{x}_t = \mathbf{B}^{-1}\gamma_0 + \mathbf{B}^{-1}\mathbf{\Gamma}_1\mathbf{x}_{t-1} + \mathbf{B}^{-1}\boldsymbol{\varepsilon}_t,$$

$$\mathbf{x}_t = \mathbf{a}_0 + \mathbf{A}_1\mathbf{x}_{t-1} + \mathbf{e}_t.$$

This is referred to as the **reduced form** VAR.

Reduced Form VAR

For every SVAR, we can derive a reduced form VAR, where each equation only has **one endogenous variable**, but

$$\text{Var} \begin{bmatrix} e_{yt} \\ e_{zt} \end{bmatrix} = \underbrace{\begin{pmatrix} 1 & b_{12} \\ b_{21} & 1 \end{pmatrix}^{-1}}_{\mathbf{B}^{-1}} \underbrace{\begin{pmatrix} \sigma_y^2 & 0 \\ 0 & \sigma_z^2 \end{pmatrix}}_{\Sigma} \underbrace{\begin{pmatrix} 1 & b_{21} \\ b_{12} & 1 \end{pmatrix}^{-1}}_{(\mathbf{B}^{-1})'} = \underbrace{\begin{pmatrix} \omega_1^2 & \omega_{12} \\ \omega_{21} & \omega_2^2 \end{pmatrix}}_{\Omega}.$$

So, the reduced form errors e_{yt} and e_{zt} are correlated; often, these are also referred to as **VAR residuals** to distinguish them from the **structural** errors ε_{yt} and ε_{zt} .

However, reduced form errors are still white noise:

$$E(e_{yt}e_{y,t-s}) = 0, \quad E(e_{yt}e_{z,t-s}) = 0,$$

for $s = 1, 2, \dots$

Relationship Between VARs and SVARs

For any reduced form VAR, there are **infinitely many** SVARs: there exist other matrices $\tilde{\mathbf{B}}$ and $\tilde{\Sigma}$ such that $\tilde{\mathbf{B}}^{-1} \tilde{\Sigma} (\tilde{\mathbf{B}}^{-1})' = \Omega$.

Example:

$$\begin{pmatrix} 1 & 0.500 \\ 0.250 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 1.000 & 0 \\ 0 & 1.000 \end{pmatrix} \begin{pmatrix} 1 & 0.250 \\ 0.500 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1.6327 & -0.9796 \\ -0.9796 & 1.3878 \end{pmatrix},$$
$$\begin{pmatrix} 1 & 4.481 \\ 2.111 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 3.680 & 0 \\ 0 & 25.39 \end{pmatrix} \begin{pmatrix} 1 & 2.111 \\ 4.481 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1.6327 & -0.9796 \\ -0.9796 & 1.3878 \end{pmatrix}.$$

To derive the SVAR from a reduced form VAR, we need additional information (return to this next week).

VAR in n Variables

VAR models have as many equations as endogenous variables, which we will denote by n .

The number of equations in a VAR is known as the **dimension**.

The number of variables in a VAR is identical to the dimension. A **bivariate** VAR contains two endogenous variables and therefore two equations.

In general, an n -variate VAR with p lags is

$$\mathbf{x}_t = \mathbf{a}_0 + \sum_{j=1}^p \mathbf{A}_j \mathbf{x}_{t-j} + e_t, \quad E(e_t) = 0, \quad \text{Var}(e_t) = \mathbf{\Omega},$$

where $\mathbf{\Omega}$ is a **symmetric** and **positive definite** matrix.

Features of VARs

An important feature of the VAR is that estimation is very easy: a VAR is just a system of **Seamingly Unrelated Regressions** (SUR).

Consistent estimates of $\mathbf{a}_0$ and $\mathbf{A}_1, \dots, \mathbf{A}_p$ can be obtained by running OLS on each equation (or more efficiently, GLS on the n equations).

However, the number of free parameters in $\mathbf{a}_0$ and $\mathbf{A}_1, \dots, \mathbf{A}_p$ grows quickly with n , i.e. $k = n(1 + np)$.

n	p	k
3	1	12
3	4	39
10	1	110
10	4	410

Properties of VARs

All relevant properties of AR processes generalize to VARs.

To analyze the properties of VARs, define the lag operator:

$$\mathbf{A}(L) = \mathbf{I}_n - \mathbf{A}_1 L - \cdots - \mathbf{A}_p L^p,$$

and write the n -variate VAR(p) as

$$\mathbf{A}(L)\mathbf{x}_t = e_t.$$

The characteristic equation for the VAR(p) is

$$\det \mathbf{A}(z) = \det (\mathbf{I}_n - \mathbf{A}_1 z - \cdots - \mathbf{A}_p z^p) = 0.$$

Stationarity

Recall that the AR(1) model

$$x_t = a_0 + a_1 x_{t-1} + e_t$$

is **stationary** if the characteristic equation

$$1 - a_1 z = 0$$

has all roots greater than one in absolute value, which corresponds to $|a_1| < 1$.

Similarly for the VAR(1), stationarity corresponds to

$$\det(\mathbf{I}_n - \mathbf{A}_1 z) = 0$$

having all roots greater than one in absolute value.

Stationarity

Let z^* be a root of $\det \mathbf{A}(z)$. Then, by definition

$$\det (\mathbf{I}_n - \mathbf{A}_1 z^*) = 0,$$

which is equivalent to

$$\det \left(\frac{1}{z^*} \mathbf{I}_n - \mathbf{A}_1 \right) = 0,$$

Therefore, if z^* is a root of $\det \mathbf{A}(z)$, then $\frac{1}{z^*}$ is an **eigenvalue** of $\mathbf{A}_1$.

This means that stationarity for the VAR(1) corresponds to **all eigenvalues** of $\mathbf{A}_1$ being **less than one in absolute value**.

Note that when $\mathbf{A}_1$ has all eigenvalues less than one in absolute value $\mathbf{A}_1^h \rightarrow 0$ as $h \rightarrow \infty$.

Example of a Stationary VAR

Consider the VAR(1):

$$\begin{pmatrix} y_t \\ z_t \end{pmatrix} = \begin{pmatrix} 1.0846 & -0.7385 \\ 0.2462 & -0.0846 \end{pmatrix} \begin{pmatrix} y_{t-1} \\ z_{t-1} \end{pmatrix} + \begin{pmatrix} e_{yt} \\ e_{zt} \end{pmatrix}.$$

Is this VAR stationary?

Example of a Stationary VAR

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Is this VAR stationary?

The 10-period ahead forecast is:

$$E(\mathbf{x}_{t+10} | \mathbf{x}_t, \mathbf{x}_{t-1}, \dots) = \begin{pmatrix} 1.0846 & -0.7385 \\ 0.2462 & -0.0846 \end{pmatrix}^{10} = \begin{pmatrix} 0.4291 & -0.3219 \\ 0.1073 & -0.0805 \end{pmatrix}.$$

The 50-period ahead forecast is:

$$E(\mathbf{x}_{t+50} | \mathbf{x}_t, \mathbf{x}_{t-1}, \dots) = \begin{pmatrix} 1.0846 & -0.7385 \\ 0.2462 & -0.0846 \end{pmatrix}^{50} = \begin{pmatrix} 0.0063 & -0.0048 \\ 0.0016 & -0.0012 \end{pmatrix}.$$

The eigenvalues of $\mathbf{A}_1$ are 0.9 and 0.1.

Example of a Stationary VAR

Consider the VAR(1):

$$\begin{pmatrix} y_t \\ z_t \end{pmatrix} = \begin{pmatrix} 1.1462 & -0.1846 \\ 0.0615 & 0.8538 \end{pmatrix} \begin{pmatrix} y_{t-1} \\ z_{t-1} \end{pmatrix} + \begin{pmatrix} e_{yt} \\ e_{zt} \end{pmatrix}.$$

Is this VAR stationary?

Example of a Stationary VAR

Consider the VAR(1):

$$\begin{pmatrix} y_t \\ z_t \end{pmatrix} = \begin{pmatrix} 1.1462 & -0.1846 \\ 0.0615 & 0.8538 \end{pmatrix} \begin{pmatrix} y_{t-1} \\ z_{t-1} \end{pmatrix} + \begin{pmatrix} e_{yt} \\ e_{zt} \end{pmatrix}.$$

Is this VAR stationary?

The 10-period ahead forecast is:

$$E(\mathbf{x}_{t+10} | \mathbf{x}_t, \mathbf{x}_{t-1}, \dots) = \begin{pmatrix} 1.1462 & -0.1846 \\ 0.0615 & 0.8538 \end{pmatrix}^{10} = \begin{pmatrix} 3.1118 & -2.0724 \\ 0.6908 & -0.1694 \end{pmatrix}.$$

The 50-period ahead forecast is:

$$E(\mathbf{x}_{t+50} | \mathbf{x}_t, \mathbf{x}_{t-1}, \dots) = \begin{pmatrix} 1.1462 & -0.1846 \\ 0.0615 & 0.8538 \end{pmatrix}^{50} = \begin{pmatrix} 144.4799 & -108.3560 \\ 36.1187 & -27.0839 \end{pmatrix}.$$

The eigenvalues of $\mathbf{A}_1$ are 1.1 and 0.9.

VAR Companion Form

Consider the VAR(p):

$$\mathbf{x}_t = \mathbf{a}_0 + \mathbf{A}_1 \mathbf{x}_{t-1} + \cdots + \mathbf{A}_p \mathbf{x}_{t-p} + e_t,$$

and add the auxiliary identities

$$\mathbf{x}_{t-1} = \mathbf{x}_{t-1},$$

$$\vdots$$

$$\mathbf{x}_{t-p+1} = \mathbf{x}_{t-p+1}.$$

Putting the system together yields

$$\begin{pmatrix} \mathbf{x}_t \\ \mathbf{x}_{t-1} \\ \vdots \\ \mathbf{x}_{t-p+1} \end{pmatrix} = \begin{pmatrix} \mathbf{a}_0 \\ 0 \\ \vdots \\ 0 \end{pmatrix} + \begin{pmatrix} \mathbf{A}_1 & \mathbf{A}_2 & \cdots & \mathbf{A}_p \\ \mathbf{I}_n & & & 0 \\ & \ddots & & \vdots \\ & & \mathbf{I}_n & 0 \end{pmatrix} \begin{pmatrix} \mathbf{x}_{t-1} \\ \mathbf{x}_{t-2} \\ \vdots \\ \mathbf{x}_{t-p} \end{pmatrix} + \begin{pmatrix} e_t \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

VAR Companion Form

So any VAR(p) can be expressed as a VAR(1) with

$$\tilde{\mathbf{A}}_1 = \begin{pmatrix} \mathbf{A}_1 & \mathbf{A}_2 & \cdots & \mathbf{A}_p \\ \mathbf{I}_n & & & 0 \\ & \ddots & & \vdots \\ & & \mathbf{I}_n & 0 \end{pmatrix}.$$

This is very useful for working with VARs. For example, to assess the stationarity of a VAR(p), we only need to check the eigenvalues of $\tilde{\mathbf{A}}_1$.

The companion form can also be used to analyze properties of a univariate AR(p).

Determining the Lag Length p

Prior to estimation, we must determine the lag length p of the VAR.

The general approach is the same as with univariate AR(p):

- compute the multivariate generalizations of criteria such as the AIC and the BIC used in the univariate case;
- compute a likelihood ratio which tests the number of lags against a shorter alternative under the null hypothesis.

LR Test Example: $H_0 : p = 4$ against $H_1 : p = 8$. Rejecting the null implies 4 lags is not enough to capture the dynamics of the model and the residuals will still show autocorrelation.

Example: The Returns of 5 Asian Stock Markets

Consider daily returns of the stock markets of: Thailand, Hong Kong, Singapore, Korea, Japan.

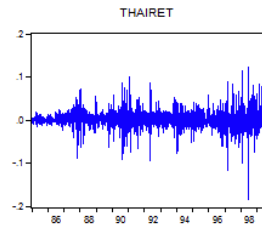
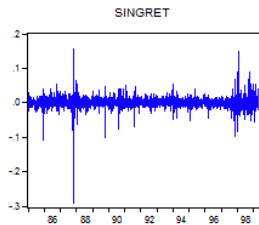
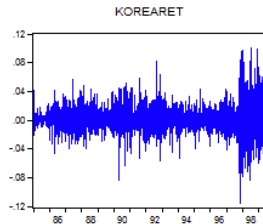
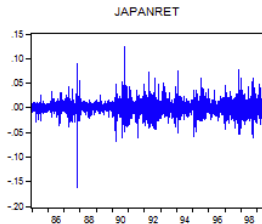
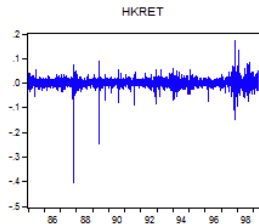
In a VAR model of these 5 stock markets we will estimate 5 equations (5 endogenous variables), one per country, with lagged values of all other stock markets appearing in each equation:

$$\mathbf{x}_t = \mathbf{a}_0 + \mathbf{A}_1 \mathbf{x}_{t-1} + \cdots + \mathbf{A}_{t-p} \mathbf{x}_{t-p} + e_t.$$

where

$$\mathbf{x}_t = (\text{THAIRET}_t, \text{HKRET}_t, \text{SINGRET}_t, \text{KOREARET}_t, \text{JAPRET}_t)'$$

Example: The Returns of 5 Asian Stock Markets



VAR Lag Length Selection

VAR Lag Order Selection Criteria

Endogenous variables: HKRET JAPANRET KOREARET SINGRET THAIRET

Exogenous variables: C

Sample: 1/01/1985 4/29/1999

Included observations: 3729

Lag	LogL	LR	FPE	AIC	SC	HQ
0	51769.00	NA	6.03e-19	-27.76294	-27.75460	-27.75997
1	52053.11	567.3073	5.25e-19	-27.90191	-27.85184*	-27.88410
2	52116.53	126.4483	5.14e-19	-27.92251	-27.83072	-27.88986
3	52191.60	149.5024	5.00e-19	-27.94937	-27.81585	-27.90187
4	52249.00	114.1630	4.92e-19	-27.96675	-27.79150	-27.90441*
5	52274.33	50.30679	4.92e-19*	-27.96693*	-27.74995	-27.88974
6	52288.08	27.27318	4.95e-19	-27.96089	-27.70219	-27.86886
7	52308.92	41.27481	4.96e-19	-27.95866	-27.65823	-27.85179
8	52331.59	44.84274*	4.96e-19	-27.95741	-27.61526	-27.83569

* indicates lag order selected by the criterion

LR: sequential modified LR test statistic (each test at 5% level)

FPE: Final prediction error